

National University of Computer and Emerging Sciences

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set of these substitutions is returned as the answer, ensuring the extracted results are logically entailed by the knowledge base.

CLO 2: Understand the challenges in representing knowledge and reasoning under uncertainty

Q2: Answer the following questions.

[Points: 5+5 =10]

- a. What is meant by Most General Unifier(MGU)? How it helps in efficient resolution process?

In simple terms, the Most General Unifier (MGU) is the simplest possible substitution that makes two logical expressions look identical. It's "most general" because it does not make any unnecessary commitments or specific assumptions. Most General Unifier is the cornerstone of efficient resolution. It is the key to making automated reasoning in first-order logic computationally feasible by ensuring the proof search remains general, flexible, and complete, while minimizing the explosive growth of derived clauses.

- b. Consider the following Knowledge Base:

Dana likes Cody
Abby does not like Dana
Everybody likes somebody
Bess likes Cody or Dana
Abby likes everyone that Bess likes.
Cody likes everyone who likes her.
Everyone likes herself.

From this KB, identifying which statement is TRUE, FALSE or UNKNOWN

1. Bess does not like Dana

TRUE

2. Everybody likes everybody

FALSE

3. Dana likes Bess.

UNKNOWN

4. Abby likes Bess.

UNKNOWN

5. Abby likes Dana.

FALSE

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CLO 4: Development of hybrid approaches by synergizing the existing framework to solve complex decision-making problems

Q3: Answer the following questions briefly to the point.

[Points: 5+5 =10]

- a. Define the term Derivation ($\vdash$). Give an example of it.

Derivation ($\vdash$) is a syntactic relationship. It states that a formula can be proven from the KB using a specific proof calculus (e.g., a set of inference rules R or reasoner).

Definition - $KB \vdash_a \alpha$ if and only if, there exists a finite sequence of wffs $\beta_1, \beta_2, \dots, \beta_n$ such that $\beta_n = \alpha$

For each i ($1 \leq i \leq n$), either: $\beta_i \in KB$, or β_i is a logical axiom (e.g., $\forall x (x = x)$), or β_i is derived from previous wffs in the sequence using an inference rule from the set R.

- b. Define the term Entailment ($\models$). Give an example of it.

Entailment is a semantic relationship between a set of formulas (the KB) and a formula (α). It is defined in terms of models.

Definition - $KB \models \alpha$ if and only if, $\forall M ((M \models KB) \rightarrow (M \models \alpha))$

Let KB be a set of well-formed formulas (wffs) in predicate logic. Let α be a wff in predicate logic. and let M be a model (an interpretation that assigns meaning to all constants, predicates, and functions).

CLO 2: Understand the challenges in representing knowledge and reasoning under uncertainty

Q4: In a Knowledge Base (KB) about scientists following knowledge items are available:

1. Every scientist who works in AI knows some mathematician.
2. Every mathematician knows every logician.
3. Anyone who knows someone that knows a logician also knows that logician.
4. Alice is a scientist who works in AI.
5. Bob is a mathematician.
6. Carol is a logician

- a. Identify the relevant Predicates for the above problem.

Constants - **Alice, Bob and Carol**

Scientist(x)- x is a scientist.

WorksInAI(x) - x works in Artificial Intelligence.

Mathematician(x)- x is a mathematician.

Logician(x) - x is a logician.

Knows(x, y) - x knows y.

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b. Convert the above knowledge items into FOL statements.

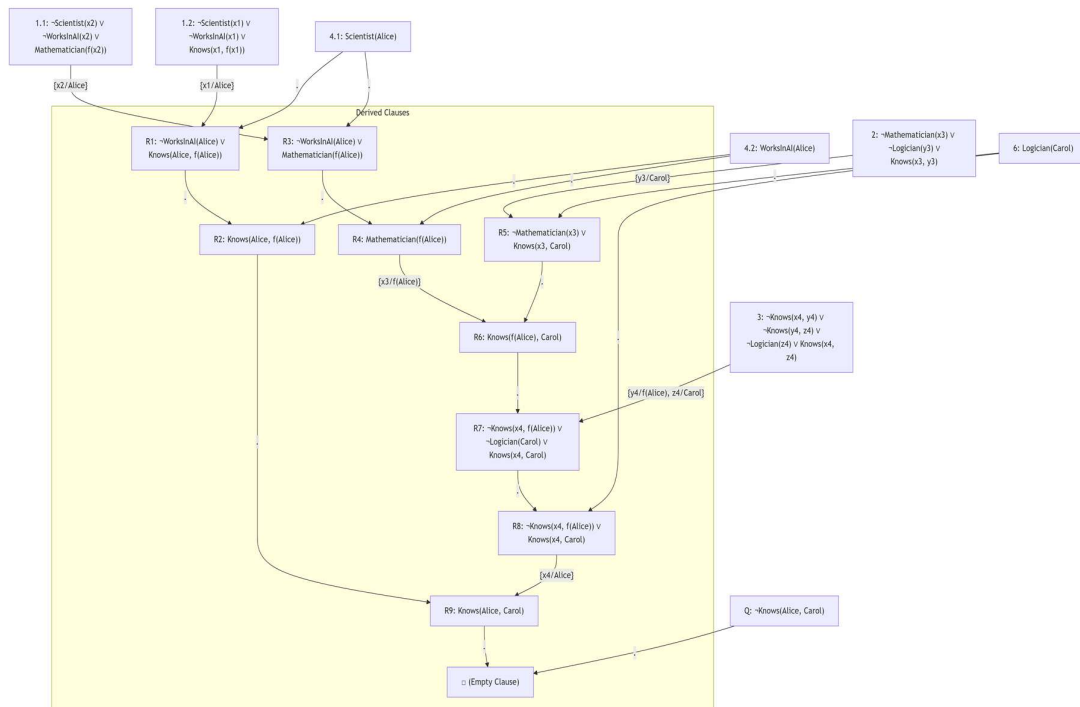
1. $\forall x [(Scientist(x) \wedge WorksInAI(x)) \rightarrow \exists y (Mathematician(y) \wedge Knows(x, y))]$
2. $\forall x \forall y [(Mathematician(x) \wedge Logician(y)) \rightarrow Knows(x, y)]$
3. $\forall x \forall y \forall z [(Knows(x, y) \wedge Knows(y, z) \wedge Logician(z)) \rightarrow Knows(x, z)]$
4. $Scientist(Alice) \wedge WorksInAI(Alice)$
5. $Mathematician(Bob)$
6. $Logician(Carl)$

To prove : $Knows(Alice, Carol)$

c. Convert all predicate logic statements into CNF form.

- 1.1. $\neg Scientist(x) \vee \neg WorksInAI(x) \vee Mathematician(f(x))$
- 1.2. $\neg Scientist(x) \vee \neg WorksInAI(x) \vee Knows(x, f(x))$
2. $\neg Mathematician(x) \vee \neg Logician(y) \vee Knows(x, y)$
3. $\neg Knows(x, y) \vee \neg Knows(y, z) \vee \neg Logician(z) \vee Knows(x, z)$
- 4.1. $Scientist(Alice)$
- 4.2. $WorksInAI(Alice)$
5. $Mathematician(Bob)$
6. $Logician(Carl)$

d. Using resolution refutation to prove that the KB entails Alice knows Carol. [Points: 2.5 X 4 =10]



<The End. >